

AIMScribe v3 — Cloud Database, or the Server's Own Disk?

What the databases actually hold, how big they really are, and which choice fits · AIMS LAB · 4 October 2026

1 · The short answer

Use the server's own SSD. Do not rent a cloud database.

The reason is one number. For all 18,000 patients — every current prescription, every previous consultation's prescription, and the complete tracking record of every audio file — **both databases together come to about 10 GB.**

That is smaller than a single one of your audio files multiplied by a hundred. It fits on a phone. A rented cloud database is built to solve problems of scale you do not have, and it would cost US\$15–30 every month to solve them.

There is one condition that flips this answer, and it has nothing to do with size or speed. It is in section 6. Read it before deciding.

2 · What the two databases actually hold

There are two, deliberately kept apart, with different logins. Neither can read the other.

aims_clinical — the patient information

What	Where it comes from
The patient: name, sex, age, phone, address	CMED, when the patient is opened
Paramedic readings: blood pressure, pulse, weight, temperature, oxygen	CMED, measured before the patient enters the room
This visit's prescription: every medicine, dose, frequency, duration	CMED, when the doctor presses Build Prescription
Diagnoses and investigations ordered	the same message
The previous consultation's prescription, with its date	CMED, as a backfill, when we do not already hold it
Every message from CMED kept word for word, exactly as sent	stored before anything is read out of it, so nothing is ever lost to a parsing mistake
A log of who read what, append-only	the system itself

aims_recordings — the audio tracking

What	Why it is kept
One row per consultation: which clinic, which doctor, which patient, when it started and ended, how long, where the audio file is	This is how any recording is ever found again
Every piece of audio: its fingerprint, its size, where it went	So a missing or altered piece can be detected
The signed chain: one entry per piece, per pause, per resume	Each entry carries the fingerprint of the one before it, so nothing can be removed or reordered without it showing
The purge receipts	The signed proof that let each laptop delete its own copy
Whether CMED described the visit, and whether it matched	An unconfirmed recording is erased after 24 hours

An append-only audit log	Who did what, never edited
No patient names, ever	By design, so somebody searching recordings cannot see who they are

Why both matter more than their size suggests. The audio files are the valuable thing, but on their own they are a folder of filenames. The databases are what turn them into a dataset: which patient, which doctor, what was prescribed, what the paramedic measured, and proof that the recording has not been altered. **Lose the databases and the audio becomes almost uninterpretable.** That is why how they are backed up matters more than where they are hosted.

3 · How big they really are

Measured, per consultation:

Database	What	Size
recordings	The session row, including the signed manifest	~25 KB
recordings	The chain, about 26 entries	~18 KB
recordings	The pieces, about 24	~7 KB
recordings	The purge receipts	~10 KB
recordings	Grant, confirmation, audit	~3 KB
clinical	CMED's two messages, kept word for word	~6 KB
clinical	Patient, visit, demographics, paramedic readings	~2 KB
clinical	Prescription, medicines, diagnoses, investigations	~2 KB
clinical	The previous visit, when it is sent	~3 KB
	Per consultation	~76 KB

For 18,000 patients	Size
The data itself	1.3 GB
With indexes, which make searching fast	2.2 GB
Our earlier bench measurement, scaled to today's settings	4.4 GB
Plan for	10 GB — generous, with room for growth and bloat

Put next to the audio, the comparison is stark. The audio for those same 18,000 patients is 2,160 GB. The databases are 10. The databases are **one half of one percent** of your storage problem — and one hundred percent of your ability to make sense of the rest.

4 · The two choices, in plain terms

Choice 1 — on the server's own SSD (how it is built today)

PostgreSQL runs in a container on the same machine as everything else. The data sits on that machine's disk. This is what the system does now and what has been tested.

Good	Less good
Costs nothing extra	Backups are your job. If nobody sets them up and tests them, there are none
Fastest possible — the database is a few microseconds away, not a network hop	If the machine dies, the database dies with it unless a backup exists elsewhere
Already built, already tested, nothing to change	Patching PostgreSQL is your job
No data ever leaves the machine, which simplifies the ethics question	Disk space is your job to watch

Choice 2 — a managed cloud database

You rent PostgreSQL from DigitalOcean, Amazon or similar. They run it, patch it, back it up and can restore it to any moment in the past few days. Your server connects to it over the network.

Good	Less good
Backups happen whether anyone remembers or not	US\$15–30 a month, for ever

Restore to any point in time, usually the last 7 days	Every query crosses a network: milliseconds instead of microseconds
Storage grows by itself — the database can never fill the disk	Patient data now sits on a third party's system, which is an ethics question
Somebody else is on call for it	One more account, one more credential, one more thing that can be unreachable

5 · Side by side, for this project

What matters here	Own SSD	Managed	Which wins
Is 10 GB too big?	No	No	neither — it is tiny for both
Speed	microseconds	1–5 ms per query	own SSD
Cost over five years	\$0	\$900 – \$1,800	own SSD
Work to switch	none — already built	a day, plus testing	own SSD
Backups	you must build and test them	automatic	managed
Recover to a point in time	only to last night's backup	any moment in the last week	managed
If the server dies	database gone unless backed up elsewhere	database survives	managed
Patient data location	stays on your machine	on a vendor's machine	own SSD
Disk can fill	yes — must be watched	no	managed

Read that table carefully and a pattern appears. Every row the managed database wins is about the same thing: **what happens when something goes wrong and nobody was watching.** Not speed, not size, not cost. You are not choosing a database. You are choosing who is responsible for backups.

6 · The recommendation, and the condition that flips it

Use the server's own SSD — because 10 GB is nothing, because it is already built and tested that way, because changing it now adds risk to a project about to start collecting, and because it keeps patient data on your own machine.

But rent a managed database instead if you cannot honestly promise this:

The promise
A nightly backup of both databases runs automatically, is encrypted, and is copied off the machine.
Somebody is told, the same day, if a backup fails.
One backup is actually restored and checked on day one, and then once a month, for as long as the study runs.

That is the whole decision. A managed database costs about \$20 a month and buys you exactly one thing: those three promises are kept by a company instead of by a person who has other work to do. **If the promises will be kept, keep the money. If they will not, \$20 a month is cheap insurance on a dataset that cannot be recreated** — those patients will not come back for a second recording.

7 · What you must do either way

	Task	Why
1	Nightly dump of both databases, encrypted, uploaded to object storage	Even a managed database should have a copy you control. Providers have lost data
2	Restore one, on day one, and prove it works	A backup nobody has read is not a backup. This is the single most skipped step in every project like this
3	Restore again once a month	Backups rot quietly: a changed password, a full bucket, an expired key
4	Alert if a backup fails	Silent failure is the normal failure

5	Keep the databases on SSD, never on a network drive	The integrity chain depends on writes landing in order. A network filesystem cannot promise that
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8 · The full picture

Everything discussed so far, in one place.

What	Where	How big	Cost
aims_clinical — patients, prescriptions, previous visits	the server's SSD	~4 GB	included
aims_recordings — audio tracking, chains, receipts	the server's SSD	~6 GB	included
The dashboard	the server	a few MB	included
Working files — joining, compressing, encrypting	the server's SSD	~3 GB	included
Offload buffer — recordings waiting to upload	the server's SSD	50 GB	included
Operating system, containers, logs, WAL, backups	the server's SSD	~60 GB	included
Raw WAV audio	object storage, rolling window	2–3 TB	~\$45/mo
FLAC copies	deep archive, permanent	6.2 TB per year	~\$6/mo, year 1

Server disk needed: 200 GB minimum, 250 GB recommended, growing by about 20–40 GB a year as the databases fill. The AIMS LAB machine's 100 GB is not enough; 8 cores and 8 GB of memory are.

Notice where the money is. The databases — the part you asked about — cost nothing and take 10 GB. **The audio is the entire cost and the entire capacity problem.** That is where the attention belongs.

9 · One page to remember

Question	Answer
Cloud database or the server's SSD?	The server's SSD. Unless nobody will test restores monthly — then rent one
How big are the databases?	About 10 GB for all 18,000 patients. Tiny
What is in them?	Patient details, paramedic readings, this visit's prescription, the previous visit's prescription, and the complete tracking record of every audio file
Why two databases?	So a person who can search recordings cannot see patient names. Separate logins, neither can read the other
What does a managed one cost?	\$15–30 a month — \$900 to \$1,800 over five years
What does it actually buy?	Backups that happen without anyone remembering. Nothing else you need
Speed difference?	The server's own disk is about a thousand times closer. Neither is too slow for this
Can the databases fill the disk?	Yes — at 20–40 GB a year. Watch it, or buy managed
Where do working files go?	The server's SSD, always. One consultation at a time, then deleted
The one step not to skip	Restore a backup on day one and prove it works
The biggest risk to the data	Not the database. It is the audio offload and whether anyone notices when it stops

Prepared by AIMS LAB. Figures measured on the project's own bench; prices indicative, October 2026.